PÓLYA INEQUALITIES FOR DISPLACED SQUARE HOLES AND PERFORATED SQUARES

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Abstract. We prove Pólya's inequality for every Dirichlet eigenvalue of a square with an arbitrary strictly interior parallel square removed. The hole may have any size and any position; neither centering nor a positive lower bound on the boundary gaps is required. The proof combines complementary square counts, endpoint-sensitive spectral gaps, and a displacement-uniform transport between the actual Dirichlet form domains. Mixed-boundary comparisons and explicit scalar estimates cover the thin-gap regime. We also prove the inequality for finitely many independently positioned and oriented square holes whose total side length is at most $\sqrt{1-12/(5\pi)}$ times the outer side. This threshold is exact for the universal square-deficit subtraction criterion, not for the geometric inequality. A further conclusion treats an arbitrary parallel rectangular hole whenever both relative side lengths are at least 11/12, uniformly in their two gap scales and in position. Analytic estimates control all unbounded spectral ranges. Exact integer and rational certificates verify the remaining real parameter intervals, without approximating eigenvalues of a perforated domain.

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1. Introduction

Weyl's law determines the leading density of the Dirichlet spectrum, but does not decide on which side of its leading term the counting function lies at a specified energy. Pólya's conjecture asks for this one-sided assertion at every energy. In the plane it states that

$$|\Omega|\lambda_k^D(\Omega) \geq 4\pi k \qquad (k \geq 1), \tag{1.1}$$

where the eigenvalues of the ordinary Euclidean Dirichlet Laplacian are repeated according to multiplicity. The coefficient is forced by Weyl's law. The difficulty is to retain it simultaneously through the low spectrum, intermediate spectral clusters, and the unbounded high-energy range.

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Perforated domains expose a basic tension in this question. Removing a hole increases Dirichlet eigenvalues, but decreases the area multiplying them in (1.1). Domain monotonicity controls only the first change. Moreover, moving a hole destroys symmetry without changing its area or its isolated spectrum. A comparison that uses the isolated spectra alone is consequently position-independent, but need not retain enough information to prove the sharp inequality. This paper combines that comparison with geometric form estimates and a transport of indexed spectral gaps. The transport constants do not depend on the displacement.

The geometric conclusions. Let Q be an open square of side $L > 0$, and let H be a closed square strictly inside Q, with sides parallel to those of Q. Our principal conclusion is

$$\boxed{|Q \setminus H|\lambda_k^D(Q \setminus H) \geq 4\pi k \qquad (k \geq 1).} \tag{1.2}$$

The side length and position of H are arbitrary. In particular, there is no centering assumption, no exceptional interval of hole sizes, and no positive lower bound on the distance between boundary components. Strict containment is required for each domain; boundary contact is not included.

For an explicit parameterization, normalize $L = 1$ and put

$$\begin{aligned} Q &= (0,1)^2, \\ H_{q,a,b} &= [(1-q)b, (1-q)b+q] \times [(1-q)a, (1-q)a+q], \\ \Omega_{q,a,b} &= Q \setminus H_{q,a,b}, \qquad 0 < q, a, b < 1. \end{aligned} \tag{1.3}$$

Thus (1.2) is the assertion $(1-q^2)\lambda_k^D(\Omega_{q,a,b}) \geq 4\pi k$ for every value of all three real parameters and every positive integer k. Similarities preserve this area-normalized assertion.

Two further conclusions describe complementary geometries. For finitely many pairwise disjoint closed squares H_i strictly inside Q, let l_i denote their side lengths. Their positions and orientations may be independent. Define

$$s_* = \sqrt{1 - \frac{12}{5\pi}} = 0.485856226839692\ldots. \tag{1.4}$$

Then

$$\sum_i l_i \leq s_* L \quad \Longrightarrow \quad \left|Q \setminus \bigcup_i H_i\right| \lambda_k^D\left(Q \setminus \bigcup_i H_i\right) \geq 4\pi k \quad (k \geq 1). \tag{1.5}$$

The finite family of holes has strictly separated boundary components. The sum-of-sides hypothesis, rather than a hypothesis on the sum of their areas, is important here.

Finally, suppose that a single closed parallel rectangle H has side lengths pL, qL and is strictly inside Q, at an arbitrary position. We obtain

$$\frac{11}{12} \leq p, q < 1 \quad \Longrightarrow \quad |Q \setminus H|\lambda_k^D(Q \setminus H) \geq 4\pi k \quad (k \geq 1). \tag{1.6}$$

The side ratios vary independently. The horizontal and vertical gap totals need not be comparable, and each positive individual gap can be arbitrarily small. The constant $11/12$ is sufficient, not asserted optimal.

The precise sharpness statement in this paper concerns the scalar comparison underlying (1.5). With

$$C(t) = \#\{(m,n) \in \mathbb{N}^2 : m^2 + n^2 \leq t^2\}, \qquad d(t) = \frac{\pi}{4}t^2 - C(t), \tag{1.7}$$

where $\mathbb{N} = \{1, 2, \ldots\}$, we prove

$$\sum_i d(q_i t) \leq d(t) \qquad \left(q_i \geq 0,\ \sum_i q_i \leq s_*,\ t \geq 0\right). \tag{1.8}$$

The number s_* is the largest total-side threshold for which this scalar assertion holds uniformly over finite lists. This is a limitation of the comparison, not an optimal geometric restriction. Indeed, (1.2) proves the geometric inequality for single parallel square holes of every size.

Prior work and the contribution here. Pólya established (1.1) for tiling planar domains [1]. The inequalities of Berezin and of Li and Yau control eigenvalue averages [2, 3]; they do not directly imply the conjectured bound at each individual index. Product constructions, packing arguments and subdivisions give other classes and reductions [4, 5, 6, 7]. Filonov, Levitin, Polterovich and Sher proved the conjecture for Euclidean balls [8] and for circular planar annuli [9]. The annulus argument combines variational bounds, special-function estimates, lattice counting and rigorous finite computation. Conformal equivalence does not transfer the ordinary Dirichlet spectrum or its sharp area-normalized bound from circular annuli to the polygonal domains considered here.

The closest comparison principle is the subtraction of complementary counting functions used by Jiang and Lin for perforated strip-tiling domains [10]. Their Theorem 1.15, in dimension two, gives the sufficient condition $\sum_i l_i/L \le 1/(2\sqrt{2}\pi)$ for square holes in a square. Their discussion also includes rectangular complementary counts and the combination of low- and high-energy estimates. We use this established variational principle. The contribution in (1.5) is the exact threshold for the universal square-deficit comparison, including the aggregation of an arbitrary finite list of hole sizes.

The principal geometric contribution is the complete family (1.2), with both displacement parameters unrestricted. Its proof supplements subtraction by an endpoint-sensitive lower bound on indexed eigenvalues and a piecewise-affine map between the actual form domains. The resulting constants are independent of the two gap fractions. For (1.6), a one-dimensional counting comparison removes one displacement parameter, while a concavity argument at low energy separates the other from two independent gap scales.

All conclusions concern unsmoothed counts at every energy. Weyl asymptotics, quantitative remainders and Riesz-mean asymptotics provide important context [11, 10, 12], but do not alone resolve the finite spectral range with the exact conjectured coefficient. Our elementary disk discrepancy estimate supplies an explicit tail. Rational interval and integer-lattice certificates settle the remaining comparisons. No approximate eigenvalues of a perforated domain enter the proof.

Organization of the argument. Section 2 develops the form comparisons and the transport of spectral gaps, with strict and inclusive counting conventions kept separate. Section 3 supplies the one-dimensional estimates used when gaps are thin. Section 4 proves the sharp total-side criterion. Section 5 joins two continuous size bands to the small- and large-hole ranges, proving (1.2). Section 6 proves the large rectangular-hole assertion independently of the disk discrepancy estimate. The appendices give the analytic discrepancy proof and every finite certificate obligation.

The finite computations are part of a proof over real parameter intervals, not a grid of sampled domains. Each accepted interval or box has a bound valid throughout it, its boundary is included with the correct spectral convention, and an analytic estimate controls the unbounded range. The scope is precisely (1.2), (1.5) and (1.6). There is no assertion for arbitrary planar domains, unrestricted large rotated holes, all rectangular holes, or Neumann eigenvalues.

2. Form comparisons and transported spectral gaps

Every domain used below is bounded and polygonal, with finitely many separated boundary components. Its Dirichlet Laplacian is the selfadjoint operator associated with

$$\mathfrak{a}_\Omega[u] = \int_\Omega |\nabla u|^2 \,\mathrm{d}x, \qquad u \in H_0^1(\Omega),$$

in $L^2(\Omega)$. The resolvent is compact, and

$$\lambda_k^D(\Omega) = \inf_{\substack{V \subset H_0^1(\Omega) \\ \dim V = k}} \sup_{0 \neq u \in V} \frac{\int_\Omega |\nabla u|^2}{\int_\Omega |u|^2}. \tag{2.1}$$

We use the inclusive counting function

$$N_\Omega(E) = \#\{k : \lambda_k^D(\Omega) \le E\}. \tag{2.2}$$

Thus $N_\Omega(E) \le |\Omega|E/(4\pi)$ at every energy implies (1.1) by evaluation at an eigenvalue. When a spectral gap gives only a lower bound on an indexed eigenvalue, we use that bound directly rather than infer an inclusive count at the gap endpoint.

Inserting the hole boundaries. Normalize the outer side to one and set

$$t = \frac{\sqrt{E}}{\pi}, \qquad e = t^2. \tag{2.3}$$

The eigenvalues of an isolated square of side q are $\pi^2(m^2+n^2)/q^2$, regardless of its position or orientation. Let P be the outer square with finitely many square holes removed. Extension by zero embeds the direct sum of the Dirichlet form domains on P and the hole interiors into the form domain on the outer square, preserving norms and energies. Min–max gives

$$N_P(E) + \sum_i C(q_i t) \le C(t). \tag{2.4}$$

In particular, (1.8) implies

$$N_P(E) \le C(t) - \sum_i C(q_i t) \le \frac{\pi}{4}\left(1 - \sum_i q_i^2\right) t^2 = \frac{|P|}{4\pi} E. \tag{2.5}$$

The inclusion of form domains, rather than an assumption on eigenfunctions at the interfaces, fixes the direction of this inequality.

Transport between actual displaced-hole operators. Fix the normalized gap fractions $a, b \in (0,1)$ in (1.3). If $0 < q \le r < 1$, map each of the three coordinate intervals (first gap, hole projection, last gap) linearly from the q configuration to the r configuration. The outer endpoints are fixed. The two gap slopes are the same, independently of their relative lengths:

$$A = \frac{1-r}{1-q} \le 1, \qquad C_0 = \frac{r}{q} \ge 1,$$

where C_0 is the slope on the middle interval. The product map Ξ is a bi-Lipschitz bijection from $\Omega_{q,a,b}$ onto $\Omega_{r,a,b}$. Its derivative is, almost everywhere, one of

$$\operatorname{diag}(A, A), \quad \operatorname{diag}(A, C_0), \quad \operatorname{diag}(C_0, A).$$

The region of type $\operatorname{diag}(C_0, C_0)$ is precisely the deleted square.

For $v \in H_0^1(\Omega_{q,a,b})$, put $u = v \circ \Xi^{-1}$. Change of variables gives

$$\int_{\Omega_{r,a,b}} |u|^2 \ge A^2 \int_{\Omega_{q,a,b}} |v|^2, \qquad \int_{\Omega_{r,a,b}} |\nabla u|^2 \le \frac{C_0}{A} \int_{\Omega_{q,a,b}} |\nabla v|^2.$$

Indeed, the transformed energy matrix is the identity on the first region and has eigenvalues $C_0/A, A/C_0$ on each of the other two. Composition is an isomorphism of the form domains. Applying (2.1) to corresponding k-dimensional spaces proves

$$\lambda_k^D(\Omega_{q,a,b}) \ge T(q,r)\lambda_k^D(\Omega_{r,a,b}), \qquad T(q,r) = \begin{cases} \dfrac{q(1-r)^3}{r(1-q)^3}, & q \le r, \\ 1, & q \ge r. \end{cases} \tag{2.6}$$

For the second case, the hole grows by inclusion when its side increases with a, b fixed: its left endpoints decrease and its right endpoints increase. Domain monotonicity proves that case. In particular, for $r \downarrow q$, monotonicity and the first case squeeze each indexed eigenvalue between $\lambda_k^D(\Omega_{q,a,b})$ and $T(q,r)^{-1}\lambda_k^D(\Omega_{q,a,b})$. This proves the right continuity used next, including at multiple eigenvalues.

A strict outer count and an inclusive hole count. For squared radii $s \geq 0$ define

$$\mathcal{C}_{\leq}(s) = C(\sqrt{s}), \qquad \mathcal{C}_{<}(s) = \#\{(m,n) \in \mathbb{N}^2 : m^2 + n^2 < s\}.$$

For $0 < r < 1$ and $\tau > 0$, put

$$D(r,\tau) = \mathcal{C}_{<}(\tau) - \mathcal{C}_{\leq}(r^2\tau). \tag{2.7}$$

This integer is nonnegative, because $r^2\tau < \tau$. The useful endpoint estimate is

$$\lambda^D_{D(r,\tau)+1}(\Omega_{r,a,b}) \geq \pi^2\tau. \tag{2.8}$$

To see why the inner count can include equality, choose $r' \in (r,1)$ and enlarge the hole to side r', keeping a, b fixed. Every inner mode counted by $\mathcal{C}_{\leq}(r^2\tau)$ then has eigenvalue strictly below $\pi^2\tau$. For all energies $E < \pi^2\tau$ sufficiently close to this endpoint, the subtraction inequality gives

$$N_{\Omega_{r',a,b}}(E) \leq \mathcal{C}_{<}(\tau) - \mathcal{C}_{\leq}(r^2\tau).$$

Consequently its $(D(r,\tau)+1)$st eigenvalue is at least $\pi^2\tau$. Let $r' \downarrow r$ and use the continuity just proved. This establishes (2.8) without discarding a multiplicity at either boundary of the comparison.

Combining the gap and transport gives, for every reference $r \in (0,1)$,

$$k \geq D(r,\tau) + 1 \quad \Longrightarrow \quad \lambda^D_k(\Omega_{q,a,b}) \geq \pi^2 T(q,r)\tau. \tag{2.9}$$

These are bounds on the actual operator. Neither the auxiliary count nor a spectrum computed numerically is being substituted for that operator.

Why interval endpoints suffice for transport. For a fixed reference r, set $J_r(q) = (1-q^2)T(q,r)$. On $q \leq r$ it is a positive constant times $q(1+q)/(1-q)^2$, whose derivative is $(1+3q)/(1-q)^3 > 0$. On $q \geq r$ it is $1-q^2$, which decreases. The expressions agree at r. Hence, on any closed interval $[l,h] \subset (0,1)$,

$$\min_{l \leq q \leq h} J_r(q) = \min\{J_r(l), J_r(h)\}. \tag{2.10}$$

This converts a rational endpoint test into a statement for every real hole size in the interval. Uniformity in a, b was already built into (2.6).

3. Square-root sums and mixed-boundary counts

The estimates for thin gaps use one-dimensional frequencies. For $x \geq 0$ and $h > 0$, write

$$A_h(x) = \sum_{\substack{j \geq 0 \\ j+1/2 \leq hx}} \sqrt{x^2 - \frac{(j+1/2)^2}{h^2}}, \qquad R_h(x) = \sum_{\substack{j \geq 1 \\ j \leq hx}} \sqrt{x^2 - \frac{j^2}{h^2}}, \qquad R = R_1. \tag{3.1}$$

Set $A_0 = R_0 = 0$. At a frequency threshold the radical is zero. The Neumann zero modes in a spectral count will be retained separately, even at equality.

The integer-point sum. Let $g_x(y) = \sqrt{x^2 - y^2}$ on $[0,x]$, and put $c_0 = \sqrt{2/27}$. For $x \geq 1$, set $j = \lfloor x \rfloor$ and $r = x - j$. Concavity gives the composite trapezoid estimate

$$R(x) \leq \int_0^j g_x(y)\,\mathrm{d}y - \frac{x}{2} + \frac{g_x(j)}{2}.$$

The chord on $[j,x]$ lies below g_x, so its integral is at least $r g_x(j)/2$. Since $g_x(j) = \sqrt{r(2x-r)} \leq \sqrt{2x}\sqrt{r}$, and the maximum of $(1-r)\sqrt{r}$ on $[0,1]$ is $2/(3\sqrt{3})$,

$$R(x) \leq \frac{\pi}{4}x^2 - \frac{x}{2} + c_0\sqrt{x} \qquad (x \geq 1). \tag{3.2}$$

At an integral x the endpoint contribution is zero. We will also use, for every $x \geq 0$, the simpler right-endpoint-sum inequality

$$R(x) \leq \int_0^x g_x(y)\,\mathrm{d}y = \frac{\pi}{4}x^2. \tag{3.3}$$

The midpoint sum, uniformly down to small widths. For $y \geq 1/2$, let w be the largest half-integer not exceeding y and put $r = y - w \in [0, 1)$. Apply the trapezoid inequality on $[1/2, w]$, use the decreasing initial half-interval, and use the chord on $[w, y]$. This gives

$$A_1(y) - \frac{\pi}{4}y^2 \leq \frac{1-r}{2}\sqrt{y^2 - w^2} \leq c_0\sqrt{y}.$$

The argument includes the case of one summand and a summand at threshold. Below $1/2$ the sum vanishes, so this upper bound remains valid. For $y \geq 3$, the inequality $c_0/\sqrt{3} \leq 4/25$ yields a bound linear in y. The remaining closed interval $[1/2, 3]$ is covered by the 123 rational intervals specified in Appendix D. Thus

$$A_1(y) \leq \frac{\pi}{4}y^2 + \frac{4}{25}y \qquad (y \geq 0). \tag{3.4}$$

Both bounds will be needed: the asymptotic one is stronger at a wide gap, while the linear one is uniform as its width shrinks. In particular, $A_h(x) = A_1(hx)/h$ gives

$$A_h(x) - \frac{\pi}{4}hx^2 \leq \min\left\{\frac{4}{25}x, c_0\sqrt{\frac{x}{h}}\right\} \qquad (h > 0). \tag{3.5}$$

One must not use the second estimate with an implicitly bounded $h^{-1/2}$ when a displacement is unrestricted.

Removing one displacement parameter. For a parallel square hole, put $\delta = 1 - q$ and $x = \delta\sqrt{E}/\pi$. The two full-width horizontal rectangles have heights $\delta a, \delta(1-a)$; the two middle side rectangles have widths $\delta b, \delta(1-b)$ and length q. Relax matching across the horizontal cuts, and relax the entire inward horizontal edge of each horizontal rectangle to Neumann. Its other three edges remain Dirichlet. On each side rectangle, the two width ends remain Dirichlet and the two length ends are Neumann. Restriction embeds the original form domain in the resulting direct sum, preserving energy and norm. These relaxed spectra therefore give an upper bound on the count.

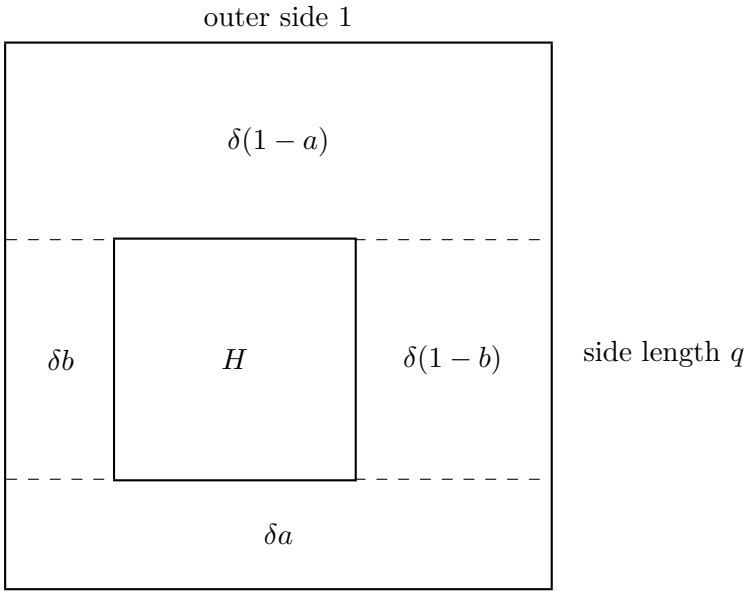


FIGURE 1. The four-rectangle comparison for a displaced square hole, with $\delta = 1 - q$. Dashed segments are artificial cuts. Matching across them is relaxed, and the entire inward edge of each full-width horizontal rectangle is also relaxed to Neumann. The hole's two vertical sides remain Dirichlet. The labels indicate widths or heights, not areas.

For the horizontal rectangles, sum the longitudinal Dirichlet counts and bound each floor by its argument. After multiplication by δ, their contribution is at most $A_a(x) + A_{1-a}(x)$. A transverse side frequency ν contributes $1 + \lfloor q\sqrt{x^2 - \nu^2}/\delta \rfloor$, where the one counts a longitudinal Neumann zero mode. For every $s \geq 0$,

$$\lfloor bs \rfloor + \lfloor (1-b)s \rfloor \leq \lfloor s \rfloor.$$

Furthermore, for $0 \leq \nu \leq x$, $\sqrt{x^2 - \nu^2} = \int_\nu^x s/\sqrt{x^2 - s^2}\,\mathrm{d}s$. Integrating the floor inequality against this nonnegative weight proves $R_b(x) + R_{1-b}(x) \leq R(x)$. Keeping the zero-mode count gives

$$\delta N_{\Omega_{q,a,b}}(E) \leq A_a(x) + A_{1-a}(x) + qR(x) + \delta \lfloor x \rfloor. \tag{3.6}$$

This step removes b without estimating a small gap by an asymptotic expression with a nonuniform constant. The same form construction, with two independent total gap widths, will be used in Section 6.

4. The exact total-side threshold for square holes

The argument for arbitrarily placed holes is entirely in terms of the deficit d in (1.7). We first construct an envelope that can aggregate any number of holes, and then retain additional partition information where that envelope is too coarse.

A deficit envelope with a monotone slope. The exact reflection identity

$$G(u) = 1 + 4\lfloor u \rfloor + 4C(u), \qquad G(u) = \#\{z \in \mathbb{Z}^2 : |z| \leq u\}, \tag{4.1}$$

and the estimate proved in Appendix A give

$$u - \frac{3}{4} - \frac{11}{4}u^{2/3} \leq d(u) \leq u + \frac{1}{4} + \frac{11}{4}u^{2/3} \qquad (u \geq 2). \tag{4.2}$$

In particular, for $u \geq 1000$,

$$\frac{2897}{4000} \leq \frac{d(u)}{u} \leq \frac{5101}{4000} < \frac{4}{3}. \tag{4.3}$$

The finite integer checks in Appendix B complete the two bounds

$$d(u) \leq \frac{4}{3}u \quad (u \geq 0), \qquad d(u) \leq \frac{1309}{1000}u \quad (0 \leq u \leq \sqrt{18}). \tag{4.4}$$

These upper bounds include left limits at every lattice jump.

Define $F(0) = 0$ and

$$F(u) = \begin{cases} \pi u^2/4, & 0 < u < \sqrt{2}, \\ \max\{\pi\sqrt{2}\,u/4,\ \pi u^2/4 - 1\}, & \sqrt{2} \leq u < \sqrt{5}, \\ (1309/1000)u, & \sqrt{5} \leq u < \sqrt{18}, \\ (4/3)u, & u \geq \sqrt{18}. \end{cases} \tag{4.5}$$

The inequality $d(u) \leq F(u)$ follows from the vanishing of C below $\sqrt{2}$, the bound $C \geq 1$ on the next range, and (4.4). Moreover, $F(u)/u$ is nonnegative and nondecreasing. In the second range it is the maximum of a constant and the increasing function $\pi u/4 - 1/u$. At $\sqrt{2}$ there is no downward jump. At $\sqrt{5}$, the left-limit prefix check in (4.4) gives $(5\pi/4 - 1)/\sqrt{5} \leq 1309/1000$, and $\pi\sqrt{2}/4 < 1309/1000$. At $\sqrt{18}$ the slope increases. It follows also that F is nondecreasing.

If $U = \sum_i u_i > 0$, this monotone slope gives

$$\sum_i d(u_i) \leq \sum_i F(u_i) \leq \sum_i u_i \frac{F(U)}{U} = F(U). \tag{4.6}$$

Zero summands and $U = 0$ cause no difficulty. Consequently $\sum_i d(q_i t) \leq F(s_* t)$ under the hypothesis in (1.8).

All but one energy band. Put $R = 243/500$. Rational enclosures of π imply

$$\frac{97}{200} < s_* < R, \qquad \frac{\pi}{4}(1 - s_*^2) = \frac{3}{5}. \tag{4.7}$$

The squared transition energies $2/s_*^2, 5/s_*^2, 18/s_*^2$ belong respectively to $(8, 9), (21, 22), (76, 77)$. Write $C_n = C(\sqrt{n})$. For the first quadratic branch of (4.5), the required inequality is $3t^2/5 \geq C(t)$; for the second quadratic branch it is $3t^2/5 \geq C(t) - 1$. The exact tests

$$3n \geq 5C_n \quad (0 \leq n \leq 8), \qquad 3n \geq 5(C_n - 1) \quad (8 \leq n \leq 21) \tag{4.8}$$

establish them. The linear branch in the maximum is covered by

$$L_n := 333n - 424C_n \geq 0, \qquad (500L_n)^2 \geq 2n(243 \cdot 333)^2 \quad (8 \leq n \leq 21). \tag{4.9}$$

Indeed, division by $\pi/4$ reduces its inequality at $t = \sqrt{n}$ to $n - 4C_n/\pi \geq s_*\sqrt{2n}$. The left side is at least $L_n/333$, and (4.9) proves the stronger bound with R in place of s_*. All quantities are checked nonnegative before squaring.

These are interval checks. On $n \leq t^2 < n + 1$, the count is C_n; the quadratic differences increase, and the linear difference has derivative $\pi t/2 - \pi\sqrt{2}s_*/4 > 0$ for $t \geq \sqrt{8}$. The overlap at $n = 8$ covers the first noninteger transition.

For the two larger branches, the exact checks give

$$\frac{d(t)}{t} \geq \frac{318087}{500000} = \frac{1309}{1000}R, \qquad \sqrt{21} \leq t < \sqrt{77}, \quad t^2 \notin [53, 54), \tag{4.10}$$

$$\frac{d(t)}{t} \geq \frac{81}{125} = \frac{4}{3}R, \qquad t \geq \sqrt{76}. \tag{4.11}$$

The finite tests proving these statements are listed in Appendix B. Between successive integer squared radii, $d(t)/t = \pi t/4 - C_n/t$ increases. The infinite part of (4.11) follows from (4.3), since $2897/4000 > 81/125$. The stated ranges overlap the transitions of F, so they prove $d(t) \geq F(s_*t)$ except possibly when $53 \leq t^2 < 54$.

Retaining the partition at the exceptional band. For every finite list $u_i \geq 0$ satisfying $\sum_i u_i \leq B := 18/5$, we have

$$\sum_i d(u_i) \leq \frac{9}{2}. \tag{4.12}$$

Here is the full argument. If every $u_i < \sqrt{2}$, then

$$\sum_i d(u_i) = \frac{\pi}{4}\sum_i u_i^2 \leq \frac{\pi\sqrt{2}}{4}B < \frac{1917}{452} < \frac{9}{2},$$

using $\pi < 355/113$ and $\sqrt{2} < 3/2$. Otherwise retain one actual member $v \geq \sqrt{2}$. Dropping the other counts and summing their quadratic costs gives

$$\sum_i d(u_i) \leq \frac{\pi}{4}\{v^2 + (B - v)^2\} - C(v), \qquad \sqrt{2} \leq v \leq B. \tag{4.13}$$

This does not suppose that only one member exceeds $\sqrt{2}$. On the constant-count intervals ending at $\sqrt{5}, \sqrt{8}, \sqrt{10}, B$, the expression is convex. The counts are respectively $1, 3, 4, 6$, since $B^2 < 13$. At an interior jump the value drops. It is therefore enough to check the initial endpoint, the three subsequent left limits, and B.

Using the positive lower radical bounds $\sqrt{2} > 7/5$, $\sqrt{5} > 223/100$, $\sqrt{8} > 14/5$, $\sqrt{10} > 79/25$, and $\pi < 355/113$, the corresponding upper bounds for (4.13) are

$$\frac{2488}{565}, \quad \frac{49973}{11300}, \quad \frac{442}{113}, \quad \frac{11349}{2825}, \quad \frac{2361}{565}. \tag{4.14}$$

All are below $9/2$. The radical bounds are verified by squaring, and each endpoint comparison is rational. This proves (4.12).

For $53 \le t^2 < 54$ the total scaled side satisfies $\sum_i q_i t \le s_* t < R\sqrt{54} < 18/5$. Meanwhile $C(t) = 37$, and

$$d(t) \ge \frac{53\pi}{4} - 37 > \frac{53}{4}\frac{333}{106} - 37 = \frac{37}{8} > \frac{9}{2}. \tag{4.15}$$

Thus (4.12) closes the entire exceptional band. Together with (4.6) and the preceding comparisons, it proves (1.8) for every real $t \ge 0$. Formula (2.5) now proves (1.5).

The exact limitation of this scalar criterion. If $S > s_*$, choose $s_* < q < \min\{S, \sqrt{2/5}\}$. This is possible because $s_* < \sqrt{2/5}$. At $t = \sqrt{5}$, the inclusive count is $C(t) = 3$, while $C(qt) = 0$. The inequality $d(t) \ge d(qt)$ would require $(5\pi/4)(1 - q^2) \ge 3$, or $q \le s_*$, a contradiction. Thus no larger threshold on the total side can guarantee (1.8). The comparison itself, rather than the geometry of the perforated domain, is what is sharp here.

5. Every parallel square hole, at every position

We now prove (1.2). The small-hole result covers $q \le s_*$, and Section 6 will cover $q \ge 11/12$. It remains to cover two closed, overlapping-endpoint bands:

$$\frac{12}{25} \le q \le \frac{4}{5}, \qquad \frac{4}{5} \le q \le \frac{11}{12}.$$

Since $12/25 < 97/200 < s_*$, the first band overlaps the small-hole range. We treat each band's unbounded energy range analytically, leaving a finite collection of indexed inequalities that the transported gaps certify uniformly in position.

A narrower disk-deficit corridor. For every $u \ge 201$, we have

$$\frac{8}{9}u \le d(u) \le \frac{10}{9}u. \tag{5.1}$$

Appendix B gives the complete finite integer tests. The explicit discrepancy estimate (4.2) supplies the infinite tail: for $u \ge 15625 = 25^3$ it gives

$$\frac{d(u)}{u} \ge \frac{27811}{31250} > \frac{8}{9}, \qquad \frac{d(u)}{u} \le \frac{17344}{15625} < \frac{10}{9}.$$

The finite part counts all positive lattice pairs through squared radius 15625^2, in bounded-memory segments. It is not a selection of radii at which the desired inequality happens to hold. The tests include inclusive lower values and the left limits needed for upper bounds at lattice jumps.

If $q \le 4/5$ and $t \ge 201/q$, both t and qt are in (5.1), so

$$d(qt) \le \frac{10}{9}qt \le \frac{8}{9}t \le d(t).$$

Consequently square subtraction proves

$$N_{\Omega_{q,a,b}}(E) \le \frac{1 - q^2}{4\pi}E \quad \left(q \le \frac{4}{5},\ E \ge \pi^2(201/q)^2\right), \tag{5.2}$$

for every displacement. No estimate of an ellipse count is required in this band.

A width-scale tail uniform in displacement. For the second band, let $\delta = 1 - q$ and $x = \delta\sqrt{E}/\pi$. Subtracting the desired Weyl term from (3.6) reduces the question to

$$\begin{aligned} D_q(x, a) = {} & A_a(x) + A_{1-a}(x) + qR(x) + (1 - q)\lfloor x \rfloor \\ & - \frac{\pi}{4}(1 + q)x^2 \le 0. \end{aligned} \tag{5.3}$$

At fixed x, a, its derivative in q is $R(x) - \lfloor x \rfloor - \pi x^2/4 \le 0$, by (3.3). It is enough to prove the inequality at $q = 4/5$, without discretizing the hole-size parameter.

For $z > 0$, define the normalized sums

$$\Phi(z) = \sum_{\substack{j\geq 0 \\ j+1/2\leq z}} \sqrt{1-((j+1/2)/z)^2}, \qquad \Psi(z) = \sum_{\substack{m\geq 1 \\ m\leq z}} \sqrt{1-(m/z)^2},$$

and set $\Phi(0) = \Psi(0) = 0$. Each is nondecreasing, including at activation thresholds. Put $y = x\min(a, 1-a)$, so $0 \leq y \leq x/2$. For $x > 0$,

$$\frac{D_{4/5}(x,a)}{x} = \Phi(y) + \Phi(x-y) + \frac{4}{5}\Psi(x) + \frac{1}{5}\frac{\lfloor x \rfloor}{x} - \frac{9\pi}{20}x. \tag{5.4}$$

Section 3 gives

$$\Phi(z) - \frac{\pi}{4}z \leq \min\left\{\frac{4}{25}, \frac{c_0}{\sqrt{z}}\right\} \qquad (z > 0), \tag{5.5}$$

$$\Psi(x) - \frac{\pi}{4}x \leq -\frac{1}{2} + \frac{c_0}{\sqrt{x}} \qquad (x \geq 1), \tag{5.6}$$

where $c_0 = \sqrt{2/27}$. At $z = 0$, the corresponding midpoint excess is zero.

First suppose $x \geq 32$ and $8 \leq y \leq x/2$. On this interval $y^{-1/2} + (x-y)^{-1/2}$ decreases with y. Using (5.4) and $\lfloor x \rfloor \leq x$ therefore gives

$$\frac{D_{4/5}(x,a)}{x} \leq c_0\left(\frac{1}{\sqrt{8}} + \frac{1}{\sqrt{24}} + \frac{4}{5\sqrt{32}}\right) - \frac{1}{5} < 0.$$

For an exact comparison, use $c_0 < 3/11$ and $1/\sqrt{8} < 71/200$, $1/\sqrt{24} < 41/200$, $1/\sqrt{32} < 177/1000$. Their substitution gives $2631/13750 < 1/5$.

Second, for every $0 \leq y \leq x/2$ and $x \geq 256$, use the uniform $4/25$ bound for the narrower contribution and $c_0/\sqrt{x-y} \leq c_0\sqrt{2/x}$ for the wider one. Then

$$\frac{D_{4/5}(x,a)}{x} \leq -\frac{1}{25} + \frac{c_0(\sqrt{2}+4/5)}{\sqrt{x}} < 0. \tag{5.7}$$

For example, $c_0 < 3/11$, $\sqrt{2} < 10/7$ and $\sqrt{x} \geq 16$ make the right side less than $-31/15400$. This estimate is uniform as either gap fraction tends to zero.

Only the closed rectangle

$$32 \leq x \leq 256, \qquad 0 \leq y \leq 8 \tag{5.8}$$

remains. Certifying this rectangle, including points beyond $y \leq x/2$ if any, is sufficient. Here in fact $y \leq 8$ and $x \geq 32$ already imply that restriction. Monotonicity of Φ, Ψ gives exact upper bounds on full rational boxes. Appendix C describes the integer test and the complete subdivision. It accepts 865759 covering boxes, with none left unresolved. Thus (5.3) holds for $q \geq 4/5$, $x \geq 32$, all real gap fractions, and hence

$$N_{\Omega_{q,a,b}}(E) \leq \frac{1-q^2}{4\pi}E \quad \left(q \geq \frac{4}{5},\ E \geq \pi^2\frac{32^2}{(1-q)^2}\right). \tag{5.9}$$

Reducing a high-energy count to finitely many indices. Fix a closed size interval $[l, h] \subset (0, 1)$ within one of the two bands, and let e_0 be a squared-energy onset valid for every size in it. We use

$$e_0 = \begin{cases} (201/l)^2, & [l,h] \subset [12/25, 4/5], \\ 32^2/(1-h)^2, & [l,h] \subset [4/5, 11/12]. \end{cases} \tag{5.10}$$

Let $\pi_- = 103993/33102 < \pi < 104348/33215 = \pi_+$; the rational enclosures are certified in Appendix B. Put

$$K(l,h) = \left\lfloor \frac{\pi_+}{4}(1-l^2)e_0 \right\rfloor. \tag{5.11}$$

Every index $k > K(l,h)$ already satisfies the desired eigenvalue inequality. Indeed, $4\pi k/(1-q^2) > \pi^2 e_0$ for $q \in [l,h]$. If $\lambda_k^D(\Omega_{q,a,b})$ were smaller than that target, an energy strictly between the target and $\max\{\lambda_k^D(\Omega_{q,a,b}), \pi^2 e_0\}$ would have count at least k, whereas the tail bound

would make it strictly less than k. This contradiction also deals correctly with an eigenvalue equal to the onset energy.

Certifying the remaining indexed inequalities. For integer $\tau \geq 1$, (2.7) becomes

$$D(r,\tau) = C_{\tau-1} - C_{\lfloor r^2\tau \rfloor}, \qquad C_n = C(\sqrt{n}).$$

For each rational reference r, keep the largest available integer $\tau \leq 250000$ whose gap index is at most k:

$$B_r(k) = \max\big(\{\tau \in \mathbb{N} : \tau \leq 250000, D(r,\tau) + 1 \leq k\} \cup \{0\}\big). \tag{5.12}$$

This finite maximum need not be the optimal operator bound; every candidate in it is justified by (2.9). With $S = 2^{40}$ define

$$\mu(l,h;r) = \left\lfloor \frac{S\pi_-}{4} \min\{(1-l^2)T(l,r), (1-h^2)T(h,r)\} \right\rfloor. \tag{5.13}$$

The entirely integer test

$$\mu(l,h;r)B_r(k) \geq Sk \tag{5.14}$$

proves $(1-q^2)\lambda_k^D(\Omega_{q,a,b}) \geq 4\pi k$ for every $q \in [l,h]$ and every $a,b \in (0,1)$. This follows by multiplying (2.9) by $(1-q^2)/(4\pi)$, using (2.10), and rounding only in the direction that weakens the lower bound. The certificate is thus on a full interval of geometries, not at its endpoints alone.

The accompanying rational reference data cover the two bands by adjoining closed intervals. Default references verify most indices; the remaining ones have explicit additional reference gaps. Every such reference, its count and its integer comparison are checked anew by the verifier. The full totals are

Size band	Intervals	Index tests	Extra gaps	Largest K
$[12/25, 4/5]$	582	24288264	130	105989
$[4/5, 11/12]$	739	9638177	81	18515

Each index test refers to an entire size interval and all displacements. There are no unverified intervals or indices. The construction and independent verification obligations are specified in Appendix C.

The finite indexed bounds and the two analytic tails now prove (1.2) on $[12/25, 11/12]$. Section 4 supplies every $q \leq s_*$, and Section 6 supplies every $q \geq 11/12$. The overlaps include all endpoints. This completes the argument for every $0 < q < 1$, every strictly interior position and every eigenvalue index.

6. Large parallel rectangular holes with arbitrary displacement

The last geometric conclusion allows the two sides of the hole to vary independently. Let the outer square have side one and let the strictly interior parallel rectangle have side lengths $p, q \in [11/12, 1)$. Its position is arbitrary. We prove

$$N_\Omega(E) \leq \frac{1-pq}{4\pi} E \qquad (E \geq 0).$$

Scaling and evaluation at each eigenvalue then give (1.6). This argument does not use the disk-discrepancy estimate.

Rotate the whole configuration by a quarter turn, if necessary, so that $p \leq q$, with p horizontal. Put

$$\delta = 1 - q, \qquad c = \frac{1-p}{1-q} \geq 1, \qquad x = \delta\frac{\sqrt{E}}{\pi}, \qquad z = cx.$$

The two horizontal gaps have lengths δa and $\delta(1-a)$, and the side gaps have lengths $c\delta b$ and $c\delta(1-b)$, for independent $a, b \in (0,1)$. The width-indexed sums A_h, R_h and $R = R_1$ are those defined in (3.1).

Cut the frame into its two full-width horizontal rectangles and two middle side rectangles. Relax matching at the cuts and the entire inward horizontal edge of each horizontal rectangle

to Neumann. Its other three edges remain Dirichlet. The side rectangles are Dirichlet across their widths and Neumann at both ends of their length q. This enlarges the original Dirichlet form domain, with norm and energy unchanged on restrictions.

The horizontal transverse frequencies are $(j+1/2)/(\delta a)$ and $(j+1/2)/(\delta(1-a))$; summing their longitudinal Dirichlet counts and bounding floors gives $\delta^{-1}(A_a + A_{1-a})$. Each side transverse frequency contributes one plus a floor, with the one counting the longitudinal Neumann zero mode even at equality. For every $s \geq 0$,

$$\lfloor bs \rfloor + \lfloor (1-b)s \rfloor \leq \lfloor s \rfloor.$$

Integration against $s/\sqrt{x^2-s^2}$ gives $R_b(x) + R_{1-b}(x) \leq R(x)$, because a frequency $\nu \leq x$ contributes $\sqrt{x^2-\nu^2}$. Applying this comparison to total side width c, and retaining the zero modes, yields

$$\delta N_\Omega(E) \leq A_a(x) + A_{1-a}(x) + \frac{q}{c} R(cx) + \delta \lfloor cx \rfloor. \tag{6.1}$$

Here $R_c(x) = R(cx)/c$. Since $1 - pq = \delta(1+qc)$, the desired Weyl expression, multiplied by δ, is $\pi(1+qc)x^2/4$.

An extremal reduction of the side lengths. For fixed c, x, a, let

$$D = A_a(x) + A_{1-a}(x) + \frac{1-\delta}{c} R(cx) + \delta \lfloor cx \rfloor - \frac{\pi}{4}(1 + (1-\delta)c)x^2. \tag{6.2}$$

The decreasing right-endpoint sum satisfies $R(u) \leq \pi u^2/4$. Consequently

$$\frac{\partial D}{\partial \delta} = \lfloor cx \rfloor + \frac{1}{c}\left(\frac{\pi}{4}(cx)^2 - R(cx)\right) \geq 0.$$

The condition $p \geq 11/12$ gives $\delta \leq 1/(12c)$. It therefore suffices to take

$$\delta = \frac{1}{12c}, \qquad p = \frac{11}{12}, \qquad q = 1 - \frac{1}{12c}. \tag{6.3}$$

This is a scalar comparison at fixed x; no geometric inclusion is asserted.

For $x > 0$ and $z = cx \geq x$, put

$$\begin{aligned} d_R(z) &= \frac{\pi}{4} z^2 - R(z), \\ \mathcal{H}(x,a) &= \frac{A_a(x) + A_{1-a}(x)}{x} - \frac{\pi}{4} x, \\ \mathcal{K}(z,x) &= \left(1 - \frac{x}{12z}\right)\frac{d_R(z)}{z} - \frac{\lfloor z \rfloor}{12z}, \\ \mathcal{L}(z) &= \frac{11}{12}\frac{d_R(z)}{z} - \frac{\lfloor z \rfloor}{12z}. \end{aligned} \tag{6.4}$$

In (6.3), $D/x = \mathcal{H} - \mathcal{K}$. Moreover $d_R \geq 0$ and $x \leq z$ give

$$\mathcal{K}(z,x) \geq \mathcal{L}(z). \tag{6.5}$$

Scalar estimates with explicit tails. Section 3 proved the uniform midpoint estimate (3.4), with its finite certificate in Appendix D. The scaling identity $A_h(x) = A_1(hx)/h$ now gives

$$\mathcal{H}(x,a) \leq \frac{8}{25} \qquad (x > 0,\ 0 \leq a \leq 1). \tag{6.6}$$

The integer-point version of the same trapezoid argument is (3.2); its initial endpoint contributes $-y/2$. Using that estimate and $\lfloor z \rfloor \leq z$ gives

$$\mathcal{L}(z) \geq \frac{3}{8} - \frac{11}{12}\sqrt{\frac{2}{27}}\, z^{-1/2} \qquad (z \geq 1). \tag{6.7}$$

The finite certificates in Appendix D and this tail prove

$$\mathcal{L}(z) \geq \frac{7}{25} \quad (z \geq 1), \qquad \mathcal{L}(z) \geq \frac{8}{25} \quad (z \geq 4). \tag{6.8}$$

Their finite ranges are $[1,7)$ and $[4,21)$, respectively. For the two tails, the exact positive differences are

$$7\left(\frac{3}{8}-\frac{7}{25}\right)^2-\left(\frac{11}{12}\right)^2\frac{2}{27}=\frac{9061}{9720000},$$

and

$$21\left(\frac{3}{8}-\frac{8}{25}\right)^2-\left(\frac{11}{12}\right)^2\frac{2}{27}=\frac{12463}{9720000}.$$

A further compact certificate gives

$$\mathcal{H}(x,a)\le\frac{4}{15}\qquad(2\le x\le 4,\ 0\le a\le 1). \tag{6.9}$$

Concavity removes displacement at low energies. For $1\le x\le 2$ there is a more precise structural bound:

$$A_a(x)+A_{1-a}(x)\le\max\{A_1(x),\,2\sqrt{x^2-1}\}. \tag{6.10}$$

If only one width contributes, monotonicity in width bounds its sum by $A_1(x)$. If both contribute nonzero terms, a second frequency in either width would require total scaled width strictly greater than $1/2+3/2=2$. Hence each width contributes only its first nonzero term. The function $h\mapsto\sqrt{x^2-1/(4h^2)}$ is concave on its active interval. The symmetric sum is therefore maximized at $a=1/2$, giving the second term in (6.10). Additional terms at $x=2$ have value zero, so the endpoint is included.

Write

$$h_0(x)=2\sqrt{1-x^{-2}}-\frac{\pi}{4}x\qquad(1\le x\le 2).$$

Then (3.4) and (6.10) imply $\mathcal{H}\le\max\{4/25,h_0\}$. The last finite certificate is the explicit comparison

$$h_0(x)\le\mathcal{K}(z,x)\qquad(1\le x\le 2,\ x\le z<4). \tag{6.11}$$

Only the two spectral scales remain; no displacement variable occurs in this certificate.

Completion of the geometric proof. If $z\ge 4$, equations (6.6), (6.8) and (6.5) give $\mathcal{H}\le\mathcal{K}$. Suppose $z<4$, so $x\le z<4$. For $2\le x<4$, use $\mathcal{H}\le 4/15<7/25\le\mathcal{L}\le\mathcal{K}$. For $1\le x\le 2$, (6.11) treats the h_0 branch, while $4/25<7/25\le\mathcal{K}$ treats the other branch.

If $0<x<1$, both horizontal widths cannot contribute nonzero terms, so $A_a+A_{1-a}\le A_1(x)$ and $\mathcal{H}\le 4/25$. When $z\ge 1$, the lower bound $\mathcal{K}\ge 7/25$ finishes the comparison. If $z<1$, both $R(z)$ and $\lfloor z\rfloor$ vanish, giving

$$\mathcal{K}(z,x)=\frac{\pi}{4}z-\frac{\pi}{48}x\ge\frac{11\pi}{48}x.$$

For $x<1/2$ one has $\mathcal{H}=-\pi x/4<0$; for $1/2\le x<1$ one has $\mathcal{K}\ge 11\pi/96>4/25$. The case $x=0$ is the zero-energy count and requires no division. The stated closed ranges cover $x=1/2,1,2,4$ and $z=1,4$; the other integer z boundaries are covered in the finite certificate by the interval beginning there.

Thus $D\le 0$ at every energy and for every allowed position and pair of side lengths. Equation (6.1) proves the required inclusive count inequality and hence (1.6). This rectangular-hole conclusion does not cover arbitrary intermediate pairs of side lengths, large rotated holes, or an arbitrary outer rectangle.

Appendix A. An explicit disk-discrepancy estimate

We supply the analytic input to (4.2), including its constant and onset. The bound is deliberately elementary; no improvement in the Gauss-circle discrepancy exponent is claimed. With the inclusive count G from (4.1), we need

$$|G(r)-\pi r^2|\le 11r^{2/3}\qquad(r\ge 2). \tag{A.1}$$

The Bessel bound. The classical integral representation and large-argument asymptotics for J_1 will be used in their usual real normalization [13, §§10.9, 10.17]. They imply

$$|J_1(x)| \le \min\{x/2, x^{-1/2}\} \qquad (x > 0), \tag{A.2}$$

as follows. Poisson's representation is

$$J_1(x) = \frac{x}{\pi} \int_{-1}^{1} e^{ixt} \sqrt{1-t^2}\, \mathrm{d}t,$$

which gives the bound $x/2$. It also gives the second bound for $0 < x \le 1$. For $1 \le x \le 2$, the alternating power series has decreasing terms. Its lower two-term truncation is nonnegative, and its upper three-term truncation gives

$$0 \le J_1(x) \le \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} \le \frac{x}{2} - \frac{5x^3}{96} \le \frac{4}{3\sqrt{5}} < \frac{3}{5}.$$

Since $(3/5)\sqrt{2} < 1$, this proves the remaining bound on that interval.

For $x \ge 2$, put $u(x) = \sqrt{x}\, J_1(x)$ and $b(x) = 1 - 3/(4x^2)$. The Bessel equation reads $u'' + bu = 0$. Consequently

$$\mathcal{E}(x) = u'(x)^2 + b(x)u(x)^2, \qquad \mathcal{E}'(x) = b'(x)u(x)^2 \ge 0.$$

The standard function and derivative asymptotics give $\lim_{x\to\infty} \mathcal{E}(x) = 2/\pi$. Therefore $b(x)u(x)^2 \le 2/\pi$. Since $b(x) \ge 13/16$,

$$|J_1(x)|^2 \le \frac{32}{13\pi x} < \frac{1}{x},$$

completing (A.2).

Smoothing a disk without losing its boundary points. Use the Fourier transform $\widehat{f}(\xi) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i x\cdot\xi}\, \mathrm{d}x$. For the indicator χ_R of a disk of radius R,

$$\widehat{\chi}_R(\xi) = \frac{R}{|\xi|} J_1(2\pi R|\xi|) \qquad (\xi \ne 0).$$

Let $\mu_\varepsilon = \chi_\varepsilon/(\pi\varepsilon^2)$, where $0 < \varepsilon < r$. The pointwise convolution comparisons give

$$\sum_{m\in\mathbb{Z}^2} (\chi_{r-\varepsilon} * \mu_\varepsilon)(m) \le G(r) \le \sum_{m\in\mathbb{Z}^2} (\chi_{r+\varepsilon} * \mu_\varepsilon)(m). \tag{A.3}$$

The lower convolution is zero at an exterior tangent point; the upper convolution equals one throughout the closed radius-r disk. Hence the inclusive convention in G introduces no boundary error.

Write $c = 1/(\pi\sqrt{2\pi})$. From (A.2),

$$|\widehat{\chi}_R(m)| \le \sqrt{\frac{R}{2\pi}} |m|^{-3/2}, \qquad |\widehat{\mu}_\varepsilon(m)| \le \min\{1, c(\varepsilon|m|)^{-3/2}\}.$$

There are exactly $8k$ points of $\mathbb{Z}^2$ with supremum norm k, and their Euclidean norm is at least k. Put $K = c^{2/3}/\varepsilon = 1/(2^{1/3}\pi\varepsilon)$. The relevant summand is decreasing, so

$$\begin{aligned}
&\sum_{m\ne 0} |m|^{-3/2} \min\{1, c(\varepsilon|m|)^{-3/2}\} \\
&\quad \le 8 \sum_{k\ge 1} k^{-1/2} \min\{1, (K/k)^{3/2}\} \le 8 \int_0^\infty y^{-1/2} \min\{1, (K/y)^{3/2}\}\, \mathrm{d}y = 24\sqrt{K}.
\end{aligned} \tag{A.4}$$

The Fourier series of the periodized convolution is absolutely convergent: the product decay is $O(|m|^{-3})$. The convolution is continuous and compactly supported. Periodization and uniqueness of its absolutely convergent Fourier series therefore justify Poisson summation here. Its nonzero frequencies have total absolute value at most

$$\frac{24}{2^{2/3}\pi} \sqrt{R/\varepsilon}.$$

Combining both sides of (A.3), and using $R = r + \varepsilon$ for a common bound, yields

$$|G(r) - \pi r^2| \le 2\pi r\varepsilon + \pi\varepsilon^2 + \frac{24}{2^{2/3}\pi}\sqrt{(r+\varepsilon)/\varepsilon}. \tag{A.5}$$

Choose $\varepsilon = \frac{1}{2}r^{-1/3}$. After division by $r^{2/3}$, the right side is

$$\pi + \frac{\pi}{4}r^{-4/3} + \frac{24}{2^{2/3}\pi}\sqrt{2 + r^{-4/3}}.$$

For $r \ge 2$, use $r^{-4/3} < 2/5$, $\pi < 22/7$, $\pi > 157/50$, $2^{2/3} > 19/12$, and $\sqrt{12/5} < 31/20$. The expression is less than

$$\frac{121}{35} + \frac{22320}{2983} = \frac{1142143}{104405} < 11. \tag{A.6}$$

All radical comparisons follow by taking the appropriate positive powers. This proves (A.1) with the explicit onset $r = 2$.

Finally (4.1) gives

$$d(u) = \lfloor u \rfloor + \frac{1}{4} - \frac{G(u) - \pi u^2}{4}.$$

The inequalities $u - 1 \le \lfloor u \rfloor \le u$ now give (4.2), including its different constants $-3/4$ and $+1/4$.

Appendix B. Exact arithmetic and the disk-deficit certificates

This appendix specifies the finite arithmetic inputs to Sections 4 and 5. The analytic reductions in those sections and Appendix A are part of the proof; no finite enumeration alone asserts an all-energy conclusion.

Rational constants and inclusive lattice counts. The Machin identity

$$\pi = 16\arctan(1/5) - 4\arctan(1/239)$$

and the alternating series

$$\arctan(1/a) = \sum_{j\ge 0} \frac{(-1)^j}{(2j+1)a^{2j+1}}$$

give rational enclosures. An even partial sum is a lower bound, and adding the next positive term gives an upper bound. Twelve terms for each arctangent suffice for both

$$\frac{333}{106} < \pi < \frac{355}{113}, \qquad \pi_- := \frac{103993}{33102} < \pi < \frac{104348}{33215} =: \pi_+. \tag{B.1}$$

For completeness, tangent addition gives $\tan(4\arctan(1/5) - \arctan(1/239)) = 1$ exactly. The same series bounds place that angle between zero and one, hence in $(0, \pi/2)$, fixing the branch $\pi/4$.

For an integer cutoff M, form

$$r_n = \#\{(a,b) \in \mathbb{N}^2 : a^2 + b^2 = n\}, \qquad C_n = \sum_{j=0}^{n} r_j \quad (0 \le n \le M). \tag{B.2}$$

Enumerating unordered pairs and assigning weight two off the diagonal and one on it retains all multiplicities. Alternatively, enumerate every ordered pair directly. The identity

$$C_n = \sum_{a=1}^{\lfloor\sqrt{n}\rfloor} \left\lfloor \sqrt{n - a^2} \right\rfloor \tag{B.3}$$

provides direct row checks. All root floors here are exact integer square roots. The small-hole verifier uses $M = 1000000$, for which $C_M = 784387$.

Upper bounds use left limits; lower bounds use inclusive values. Between consecutive integer squared radii, $d(u)/u = \pi u/4 - C_n/u$ increases. To bound it above, the finite tests therefore use left limits at the next possible jump. Put $U_n = 355n - 452C_{n-1}$. Whenever $U_n > 0$, check

$$9U_n^2 \le 1808^2 n \qquad (1 \le n \le 1000000), \tag{B.4}$$

$$(1000U_n)^2 \le (1309 \cdot 452)^2 n \qquad (1 \le n \le 18). \tag{B.5}$$

If $U_n \le 0$ the upper comparison is immediate, without squaring. These tests establish the finite parts of (4.4); (4.3) covers every $u \ge 1000$, including the last endpoint.

For lower bounds put $L_n = 333n - 424C_n$, now using the inclusive count. In addition to (4.8) and (4.9), check

$$L_n \ge 0, \quad (500000L_n)^2 \ge (318087 \cdot 424)^2 n \qquad (21 \le n \le 76,\ n \ne 53), \tag{B.6}$$

$$L_n \ge 0, \quad (125L_n)^2 \ge (81 \cdot 424)^2 n \qquad (76 \le n \le 1000000). \tag{B.7}$$

The sole failure of the middle test with $n = 53$ included is exactly that index. The partition argument (4.12)–(4.15) supplies its whole energy band. It is not omitted on numerical grounds. Monotonicity between jumps, the overlapping transition intervals in Section 4, and (4.3) give the full real ranges.

The program `verify_deficits.py` reconstructs these counts and tests, the rational bounds placing s_* and its three transitions, the five partition endpoint bounds, and the numerical constants in Appendix A. It takes no certificate data as input. Its arithmetic is standard-library integer and rational arithmetic.

The corridor beginning at radius 201. The program `verify_corridor.py` constructs the complete positive-quadrant count through $M = 15625^2 = 244140625$. Frequencies are accumulated in segments of at most 4000000 unsigned 32-bit entries; cumulative counts carry between adjacent segments. Thus the frequency segment occupies at most 16 MB, not an array proportional to the full cutoff.

On $n \le u^2 < n+1$, the lower and upper endpoint tests for (5.1) are the conservative integer inequalities

$$9(103993n - 4 \cdot 33102C_n) \ge 32 \cdot 33102\lceil\sqrt{n}\rceil, \tag{B.8}$$

$$9(104348(n+1) - 4 \cdot 33215C_n) \le 40 \cdot 33215\lfloor\sqrt{n+1}\rfloor. \tag{B.9}$$

The first tests the inclusive lower endpoint, rounding the required right side upward. The second tests the left limit at the upper endpoint, rounding its permissible right side downward. The use of rational lower and upper bounds for π has the same conservative directions.

The last lower failure occurs at $n = 17226$, and the last upper failure at $n = 40320$. Both are below $201^2 = 40401$. The final count is $C_{244140625} = 191731890$. All subsequent finite bands pass. The analytic tail in Section 5 begins at radius 15625, so there is no unverified endpoint between the finite and infinite parts. The verifier explicitly requires the certified onset to be at most 201.

For speed, this and the thin-tail verifier use NumPy integer arrays. A floating square root is used only to propose an integer r; the inequalities $r^2 \le n < (r+1)^2$ are then checked exactly before the integer root is used. Failure of that validation stops verification. Floating-point tolerances do not decide any proof inequality. Array sizes, cumulative counts and all fixed-width integer products are bounded before use; scalar rational calculations use arbitrary-precision Python integers.

Appendix C. Interval certificates for displaced square holes

Two different finite tasks occur in the displaced-hole proof: a scalar inequality in a compact two-dimensional region, and indexed eigenvalue bounds on intervals of hole sizes. Their analytic dependencies and their acceptance tests are distinct.

The compact thin-tail rectangle. Use denominator $D = 2048$ and fixed-point scale $H = 2^{20}$. For $z = k/D > 0$ and a positive integer j, put

$$V_{k,j} = 4k^2 - D^2 j^2.$$

When this is nonnegative, $\sqrt{V_{k,j}}/(2k)$ is a summand of $\Phi(k/D)$ for odd j and of $\Psi(k/D)$ for even j. The range $1 \le j \le 512$ contains every possible summand needed for $0 \le k/D \le 256$. An exact integer-root validation supplies $U_{k,j} = \lceil \sqrt{V_{k,j}} \rceil$, with no added unit when the radical is already a perfect square. Explicitly, for $k > 0$ put

$$f_k = \left\lceil \frac{H}{2k} \sum_{\substack{1 \le j \le 512 \\ j \text{ odd},\ V_{k,j} \ge 0}} U_{k,j} \right\rceil, \qquad g_k = \left\lceil \frac{H}{2k} \sum_{\substack{1 \le j \le 512 \\ j \text{ even},\ V_{k,j} \ge 0}} U_{k,j} \right\rceil.$$

These integer arrays satisfy

$$\Phi(k/D) \le f_k/H, \qquad \Psi(k/D) \le g_k/H.$$

Set $f_0 = g_0 = 0$. The summation and division directions are important; the arrays are upper enclosures, not rounded approximations to the normalized sums.

Start with the 1792 closed unit boxes $[n, n+1] \times [m, m+1]$, $32 \le n \le 255$, $0 \le m \le 7$. For a box in integer coordinates

$$l/D \le x \le u/D, \qquad v/D \le y \le w/D,$$

monotonicity in (5.4) gives

$$\begin{aligned} 5D_{4/5}(x,a)/x \le{}& 5\Phi(w/D) + 5\Phi((u-v)/D) + 4\Psi(u/D) \\ &+ \frac{\lfloor u/D \rfloor}{l/D} - \frac{9\pi}{4}\frac{l}{D}. \end{aligned}$$

Consequently it suffices to check the integer inequality

$$\begin{aligned} 5f_w + 5f_{u-v} + 4g_u + \left\lceil \frac{H\lfloor u/D \rfloor D}{l} \right\rceil \\ - \left\lfloor \frac{9\pi_Hl}{4D} \right\rfloor \le 0. \end{aligned} \tag{C.1}$$

All positive terms have been rounded upward and the subtracted term downward. The floor value at the right endpoint includes a possible new mode, which is a safe upper bound for the entire closed box.

If a box does not pass, split it at its two midpoints into four closed children. Their union is their parent. The subdivision must stop with a verification failure if an unresolved box reaches the specified depth or integer-grid limit; it is never silently discarded. The completed run of `verify_thin_tail.py` has

tested boxes	1153748,
accepted boxes	865759,
maximum subdivision level	8,
unresolved boxes	0.

The accepted integer areas sum to $224 \cdot 8 \cdot 2048^2 = 7516192768$. Independently, the tree identity is tested $= 1792 + 4(\text{tested} - \text{accepted})$. Coverage follows from the actual subdivision procedure; the area and tree identities are additional checks, not substitutes for coverage of boundaries. The least integer slack is zero, which is valid for the non-strict inequality required here. The analytic comparisons for the two infinite regions in Section 5 are also checked as exact rational inequalities by this verifier.

The rational reference data. The two reference files are `wide_certificate.json` and `thin_certificate.json`. They specify the size intervals and reference gaps for (5.14). They are certificate inputs, not accepted proof summaries. A reference size is an integer numerator over 64000 in the first file and over 192000 in the second. Both use squared-energy cutoff 250000 and coefficient scale $S = 2^{40}$. The two denominators are not interchangeable.

For each interval $[l, h]$, the data record the cutoff $K(l, h)$, two default reference numerators, their scaled coefficients, the indices not covered by those two references, and an additional reference and integer energy for each such index. The verification does not assume the accuracy of any of these entries. The program `verify_transport.py` does the following:

(1) It certifies (B.1), binds the data to those rational constants and to the specified scales, and checks that the ordered closed intervals meet exactly, with the required first and last endpoints.
(2) It reconstructs every ordered positive lattice pair through squared radius 250000 and checks direct row counts, including $C_{250000} = 195837$.
(3) It recomputes each cutoff using (5.10)–(5.11), and every coefficient using (5.13) with exact fractions.
(4) For each default reference r, it reconstructs $B_r(k)$ from all $1 \le \tau \le 250000$ by assigning each energy to its gap index and taking cumulative maxima. Every index $1 \le k \le K(l, h)$ is tested by (5.14); the resulting uncovered list must agree exactly with the data.
(5) For each uncovered k, it verifies the additional reference's gap index directly from $C_{\tau-1} - C_{\lfloor r^2\tau \rfloor} + 1$, verifies that this index is at most k, and tests $\mu(l, h; r)\tau \ge Sk$. The additional entries must cover the uncovered list exactly once.

Thus neither a saved success flag nor the program that originally searched for references is a trusted premise. The total is 1321 intervals, 33926441 interval/index comparisons, and 211 additional gaps, with the bandwise totals displayed in Section 5.

All vectorized products are checked to fit signed 64-bit integers. For example, $2^{40} \cdot 250000 < 2^{63}$ and $192000^2 \cdot 250000 < 2^{63}$ bound the two largest types of scaled products. Coefficients are first computed with arbitrary-precision fractions; the verifier requires them to lie in $[0, 2^{40})$ before vector multiplication.

The transport verification alone supplies the finite indexed part. The corridor in Appendix B and the thin scalar certificate above supply its respective high-energy dependencies. Their combination with the form proofs, the small-hole argument and the large rectangular-hole argument is the proof of the complete geometric assertion. It is not an extrapolation from finitely many eigenvalue indices.

Appendix D. Exact certificates for the rectangular-hole comparison

The accompanying program `verify_rectangular.py` verifies the five finite statements in Section 6. It is independent of the other verifiers and uses only integer and rational arithmetic. Twelve-term alternating arctangent enclosures in the Machin identity recheck $333/106 < \pi < 355/113$.

Put $P = 333/106$ and $B_0 = 2^{24}$. For a nonnegative rational number $r = u/v$, define

$$U(r) = \operatorname{isqrt}\left(\left\lfloor \frac{uB_0^2}{v} \right\rfloor\right) + 1, \qquad T(r) = U(r) - 1. \tag{D.1}$$

Then $T(r)/B_0 \le \sqrt{r} < U(r)/B_0$, also when $r = 0$ or the radical is a perfect square. In the following acceptance tests, every upper sum uses these enclosures after monotonicity has moved its energy and width arguments to the indicated endpoints. All failed comparisons cause subdivision; exhaustion of a stated grid or depth is a fatal verification failure.

The midpoint bound and the lower envelopes. For (3.4), use squared energy $r = y^2$ and begin with $[1/4, 9]$. A rational interval $[l, u]$ is accepted if

$$\frac{P}{4}l + \frac{4}{25}\frac{T(l)}{B_0} \geq \operatorname{upper} A_1(\sqrt{u}). \tag{D.2}$$

Otherwise bisect it. The 123 accepted intervals have total width $35/4$, giving the full closed finite range $1/2 \leq y \leq 3$.

For a lower bound $\mathcal{L} \geq \gamma$, divide the z axis into half-open integer bands $n \leq z < n+1$. On a subinterval $[l, u)$ accept if

$$\frac{11P}{4}l^2 \geq 11 \operatorname{upper} R(u) + n + 12\gamma u. \tag{D.3}$$

This is obtained by multiplying the desired inequality by $12z$. Take $\gamma = 7/25$ and $n = 1, \ldots, 6$, or $\gamma = 8/25$ and $n = 4, \ldots, 20$. Rational bisection produces accepted widths summing to six and seventeen, respectively. Their final endpoints are covered by (6.7). At an integer boundary, the next band uses the new floor value.

The remaining horizontal range. For (6.9), let $r = x^2$ and use grid denominators $G_0 = K_0 = 2^{14}$ for r and a. The initial rectangle is $[4, 16] \times [0, 1/2]$; symmetry gives the remaining displacement range. A box $[l, u] \times [a, b]$ is accepted if

$$\frac{P}{4}l + \frac{4}{15}\frac{T(l)}{B_0} \geq \operatorname{upper}\{A_b(\sqrt{u}) + A_{1-a}(\sqrt{u})\}. \tag{D.4}$$

Both width sums increase in their width and squared energy, so the separate endpoint maxima give a safe upper bound even when they do not occur at a common displacement.

In integer coordinates (r_l, r_u, a_l, a_u) the energy coordinate is bisected when $(r_u - r_l)K_0 \geq 2(a_u - a_l)r_u$, unless only the other coordinate remains available. The accepted grid areas sum exactly to $12G_0(K_0/2) = 1610612736$.

The explicit two-scale comparison. For $n = \lfloor z \rfloor$, subtraction gives

$$\mathcal{K}(z, x) - h_0(x) = \frac{\pi}{4}z + \frac{11\pi}{48}x - \left(1 - \frac{x}{12z}\right)\frac{R(z)}{z} - \frac{n}{12z} - 2\sqrt{1 - x^{-2}}. \tag{D.5}$$

The quotient $R(z)/z$ is a sum of terms $\sqrt{1 - j^2/z^2}$, each increasing in z after activation. The factor $1 - x/(12z)$ increases in z and decreases in x. Therefore, on $[a, b] \times [c, d)$ within one fixed floor band, it suffices to check

$$P\left(\frac{c}{4} + \frac{11a}{48}\right) \geq \left(1 - \frac{a}{12d}\right)\operatorname{upper}\frac{R(d)}{d} + \frac{n}{12c} + 2\frac{U(1 - b^{-2})}{B_0}. \tag{D.6}$$

Here the normalized Riesz sum is enclosed directly by summing $U(1 - j^2/d^2)/B_0$.

Use denominator $G_2 = 2^{16}$ for both coordinates and initial rectangles $[1, 2] \times [n, n+1)$, $n = 1, 2, 3$. Intersect each box with $z \geq x$. A box with zero intersection area is discarded; all others are accepted by (D.6) or subdivided along a longest available coordinate, choosing x at a tie. The acceptance inequality proves the continuous frozen-n expression on the full closed box and hence on its clipped part within the half-open floor band. It does not substitute the old floor value at the next integer.

For a grid rectangle $[x_l, x_u] \times [z_l, z_u]$, twice its clipped grid area is computed exactly as

$$2(z_u - z_l)\max\{0, \min(x_u, z_l) - x_l\} + (v - u)(2z_u - v - u)$$

when $v > u$, with the last term omitted otherwise, where $u = \max(x_l, z_l)$ and $v = \min(x_u, z_u)$. The sums of accepted doubled areas on the three bands are $G_2^2, 2G_2^2, 2G_2^2$, exactly the corresponding initial clipped areas.

Within each fixed floor band, diagonal points $z = x$ are limits of the positive-area region above the diagonal. The finite collection of accepted closed boxes therefore covers them by continuity of the frozen-n expression. At $z = 1, 2, 3$ the band beginning there supplies its actual floor value; $z = 4$ belongs to (6.8). Shared subdivision edges are included. Thus the

area checks are supplemented by a boundary argument and are not being used alone to infer coverage of spectral jumps.

Replay totals. The deterministic results are as follows.

Finite comparison	Tested	Accepted	Depth
Midpoint, $1/2 \le y \le 3$	245	123	12
$\mathcal{L} \ge 7/25$, $1 \le z < 7$	434	220	9
$\mathcal{L} \ge 8/25$, $4 \le z < 21$	1995	1006	10
$\mathcal{H} \le 4/15$, $2 \le x \le 4$	5395	2698	18
Center comparison, $1 \le x \le 2$, $x \le z < 4$	9280	4620	18

All 8667 accepted boxes have positive rational margins. The triangle routine additionally skips 43 empty boxes. The program checks these statistics and the interval-width and grid-area identities against their exact expected values. The least margins, in table order, are

$$\frac{501827}{4445962240},\quad \frac{39188871}{22229811200},\quad \frac{3232271}{4445962240},\quad \frac{39331}{6668943360},\quad \frac{440730641}{125696244449280}.$$

These are enclosures on full intervals or boxes, not values sampled at selected parameters. All unbounded ranges were already reduced to the displayed finite regions by the analytic estimates in Section 6.

Verification uses unconditional checks rather than Python assertions that optimization flags can disable. Caches are bounded. The computations certify explicit scalar comparisons; no eigenvalue of a perforated domain is computed.

Note on AI Use

The author declares the use of consumer-grade Gemini-3 and GPT-6 models for prior-art search, exploring and developing ideas and proofs, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

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